

Analysing Smart City Bike Sharing Data using Power BI

STEP 1 — Import the dataset into Power BI

Navigator

Display Options ▾

📁 bike-stations-sharing-data.xlsx [1]

📄 Feuil1

Feuil1

number	name	address	position
2000	City Central Bike Station 1	102 Main Street, City	46.548876678
1297	City Central Bike Station 2	103 Main Street, City	43.293485736
8265	City Central Bike Station 3	104 Main Street, City	43.269318556
8065	City Central Bike Station 4	105 Main Street, City	43.267272515
1011	City Central Bike Station 5	106 Main Street, City	47.19504, -1.5
4342	City Central Bike Station 6	107 Main Street, City	43.305993453
6176	City Central Bike Station 7	108 Main Street, City	43.284775983
3322	City Central Bike Station 8	109 Main Street, City	43.312337033
1264	City Central Bike Station 9	110 Main Street, City	43.297896313
1191	City Central Bike Station 10	111 Main Street, City	43.296207670
5058	City Central Bike Station 11	112 Main Street, City	43.292800863
4293	City Central Bike Station 12	113 Main Street, City	43.296263405
1245	City Central Bike Station 13	114 Main Street, City	43.299027938
7153	City Central Bike Station 14	115 Main Street, City	43.289467608
6154	City Central Bike Station 15	116 Main Street, City	43.290329770
1002	City Central Bike Station 16	117 Main Street, City	43.296138776
1224	City Central Bike Station 17	118 Main Street, City	43.295825183
1287	City Central Bike Station 18	119 Main Street, City	43.299520975
500	City Central Bike Station 19	120 Main Street, City	48.799828, 1.5
5220	City Central Bike Station 20	121 Main Street, City	43.298991956
6160	City Central Bike Station 21	122 Main Street, City	43.286838351
7026	City Central Bike Station 22	123 Main Street, City	43.289856201
54000	City Central Bike Station 23	124 Main Street, City	48.799887, 1.5

Load

Transform Data

Cancel

STEP 1A — Check the raw data

Power Query Editor interface showing the raw data table with columns: number, name, address, position, banking, bonus. The table contains 23 rows of bike station data. The interface includes a ribbon with tabs like Home, Transform, Add Column, View, Tools, and Help. The bottom status bar indicates 12 COLUMNS, 999+ ROWS. The right sidebar shows the Query Settings for 'Feuil1' with the 'Changed Type' step applied.

STEP 2 — Check and set Data Types

	number	name	address	position	banking	bonus
	Valid 0% Error 0% Empty 0%	Valid 0% Error 0% Empty 0%	Valid 0% Error 0% Empty 0%	Valid 0% Error 0% Empty 0%	Valid 0% Error 0% Empty 0%	Valid 0% Error 0% Empty 0%
1	2297	City Central Bike Station 2	103 Main Street, City	43.293485736186526, 5.378423799723321	TRUE	FALSE
2	8265	City Central Bike Station 3	104 Main Street, City	43.269318556371154, 5.389387542314263	TRUE	FALSE
3	8065	City Central Bike Station 4	105 Main Street, City	43.267272515671856, 5.372088524255505	TRUE	FALSE
4	1011	City Central Bike Station 5	106 Main Street, City	47.19504, -1.5575	FALSE	FALSE
5	4342	City Central Bike Station 6	107 Main Street, City	43.305993453766845, 5.393678259953781	TRUE	FALSE
6	6176	City Central Bike Station 7	108 Main Street, City	43.284775983612235, 5.385593763617925	TRUE	FALSE
7	3322	City Central Bike Station 8	109 Main Street, City	43.3123370399008, 5.3687418043728	TRUE	FALSE

	status	Contract Name	Bike Stands	Available Bike Stands	Available Bikes	Last Update
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
7	FALSE OPEN	marseille	24	24	0	06-09-2022 18:02:17 +05:30
8	FALSE OPEN	marseille	20	13	7	14-09-2022 13:27:09 +05:30
9	FALSE OPEN	marseille	10	0	0	12-09-2022 11:18:28 +05:30
10	FALSE OPEN	marseille	20	2	0	03-10-2022 11:34:01 +05:30
11	FALSE OPEN	marseille	10	10	0	13-09-2022 15:41:24 +05:30
12	FALSE OPEN	marseille	14	0	0	26-09-2022 16:44:57 +05:30
13	FALSE OPEN	marseille	15	0	0	03-10-2022 12:02:43 +05:30

STEP 3 — Check for duplicate rows

	number	name	address	position	banking	bonus
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	2000	City Central Bike Station 1	102 Main Street, City	46.5488766782771, 15.6350265324683	FALSE	FALSE
2	1297	City Central Bike Station 2	103 Main Street, City	43.293485736186526, 5.378423799723321	TRUE	FALSE
3	8265	City Central Bike Station 3	104 Main Street, City	43.269318556371154, 5.389387542314263	TRUE	FALSE
4	8065	City Central Bike Station 4	105 Main Street, City	43.267272515671856, 5.372088524255505	TRUE	FALSE
5	1011	City Central Bike Station 5	106 Main Street, City	47.19504, -1.5575	FALSE	FALSE
6	4342	City Central Bike Station 6	107 Main Street, City	43.305993453766845, 5.393678259953781	TRUE	FALSE
7	6176	City Central Bike Station 7	108 Main Street, City	43.284775983612235, 5.385593763617925	TRUE	FALSE
8	3322	City Central Bike Station 8	109 Main Street, City	43.3123370399008, 5.3687418043728	TRUE	FALSE
9	1264	City Central Bike Station 9	110 Main Street, City	43.29789631239341, 5.381132215559973	TRUE	FALSE
10	1191	City Central Bike Station 10	111 Main Street, City	43.29620767061392, 5.37693066220596	TRUE	FALSE
11	5058	City Central Bike Station 11	112 Main Street, City	43.292800863545835, 5.401285286456447	TRUE	FALSE
12	4293	City Central Bike Station 12	113 Main Street, City	43.29626340517055, 5.406000803377506	TRUE	FALSE
13	1245	City Central Bike Station 13	114 Main Street, City	43.29902793826475, 5.381500046726694	TRUE	FALSE
14	7153	City Central Bike Station 14	115 Main Street, City	43.2894676088984, 5.36370990786361	TRUE	FALSE
15	6154	City Central Bike Station 15	116 Main Street, City	43.290329770206334, 5.375548341850035	TRUE	FALSE
16	1002	City Central Bike Station 16	117 Main Street, City	43.2961387765145, 5.3707713851323	TRUE	FALSE
17	1224	City Central Bike Station 17	118 Main Street, City	43.29582518327951, 5.385996090720193	TRUE	FALSE
18	1287	City Central Bike Station 18	119 Main Street, City	43.29952097558015, 5.385047448641554	TRUE	FALSE
19	500	City Central Bike Station 19	120 Main Street, City	48.799828, 1.946855	FALSE	FALSE
20	5220	City Central Bike Station 20	121 Main Street, City	43.29899195650368, 5.388799115293999	TRUE	FALSE
21	6160	City Central Bike Station 21	122 Main Street, City	43.28683835122117, 5.379450180172236	TRUE	FALSE
22	7026	City Central Bike Station 22	123 Main Street, City	43.28985620147398, 5.3595962903211387	TRUE	FALSE
23	54000	City Central Bike Station 23	124 Main Street, City	48.799887, 1.947001	FALSE	FALSE
24	8093	City Central Bike Station 24	125 Main Street, City	43.249867874570256, 5.389411648303773	TRUE	FALSE

mn profiling based on top 1000 rows

PREVIEW DOWNLOADED AT 10:41

STEP 4 — Split position into Latitude and Longitude

Split Column by Delimiter

Specify the delimiter used to split the text column.

Select or enter delimiter

Comma

Split at

☐ Left-most delimiter

☐ Right-most delimiter

☒ Each occurrence of the delimiter

► Advanced options

Quote Character

"

☐ Split using special characters

Insert special character

OK

Cancel

1.2 Latitude	1.2 Longitude	banking	bonus
Valid 100%	Valid 100%	Valid 100%	Valid 100%
Error 0%	Error 0%	Error 0%	Error 0%
Empty 0%	Empty 0%	Empty 0%	Empty 0%
46.54887668	15.63502653	FALSE	
43.29348574	5.3784238	TRUE	
43.26931856	5.389387542	TRUE	
43.26727252	5.372088524	TRUE	
47.19504	-1.5575	FALSE	
43.30599345	5.39367826	TRUE	
43.28477598	5.385593764	TRUE	
43.31233703	5.368741804	TRUE	
43.29789631	5.381132216	TRUE	
43.29620767	5.376930662	TRUE	
43.29280086	5.401285286	TRUE	
43.29626341	5.406000803	TRUE	
43.29902794	5.381500047	TRUE	
43.28946761	5.363709908	TRUE	
43.29032977	5.375548342	TRUE	
43.29613878	5.370771385	TRUE	
43.29582518	5.385996091	TRUE	
43.29952098	5.385047449	TRUE	
48.799828	1.946855	FALSE	
43.29899196	5.388799115	TRUE	
43.28683835	5.37945018	TRUE	
43.2898562	5.355962903	TRUE	
48.799887	1.947001	FALSE	
43.24986787	5.389411648	TRUE	
43.29280086	5.401285286	TRUE	

STEP 5 — Check the Contract Name column

Search

✓

(Select All)

✓

amiens

✓

besancon

✓

Bruxelles-Capitale

✓

cergy-pontoise

✓

creteil

✓

dublin

✓

jcdecauxbike

✓

lillestrom

✓

ljubljana

✓

lund

✓

luxembourg

✓

lyon

✓

maribor

✓

marseille

✓

mulhouse

✓

namur

✓

paris

⬆

⬇

⚠

List may be incomplete.

Load more

OK

Cancel

Sort Descending

Clear Sort

Clear Filter

Remove Empty

Text Filters

Search

- ☒ mestróm
- ☒ ljubljana
- ☒ lund
- ☒ luxembourg
- ☒ lyon
- ☒ maribor
- ☒ marseille
- ☒ mulhouse
- ☒ namur
- ☒ nancy
- ☒ nantes
- ☒ rouen
- ☒ santander
- ☒ seville
- ☒ toulouse
- ☒ toyama
- ☒ valence
- ☒ vilnius



List may be incomplete.

Load more

OK

Cancel

STEP 6 — Check status

 Sort Ascending

 Sort Descending

Clear Sort



Clear Filter

Remove Empty

Text Filters

Search

- ☒ (Select All)
- ☒ CLOSED
- ☒ OPEN



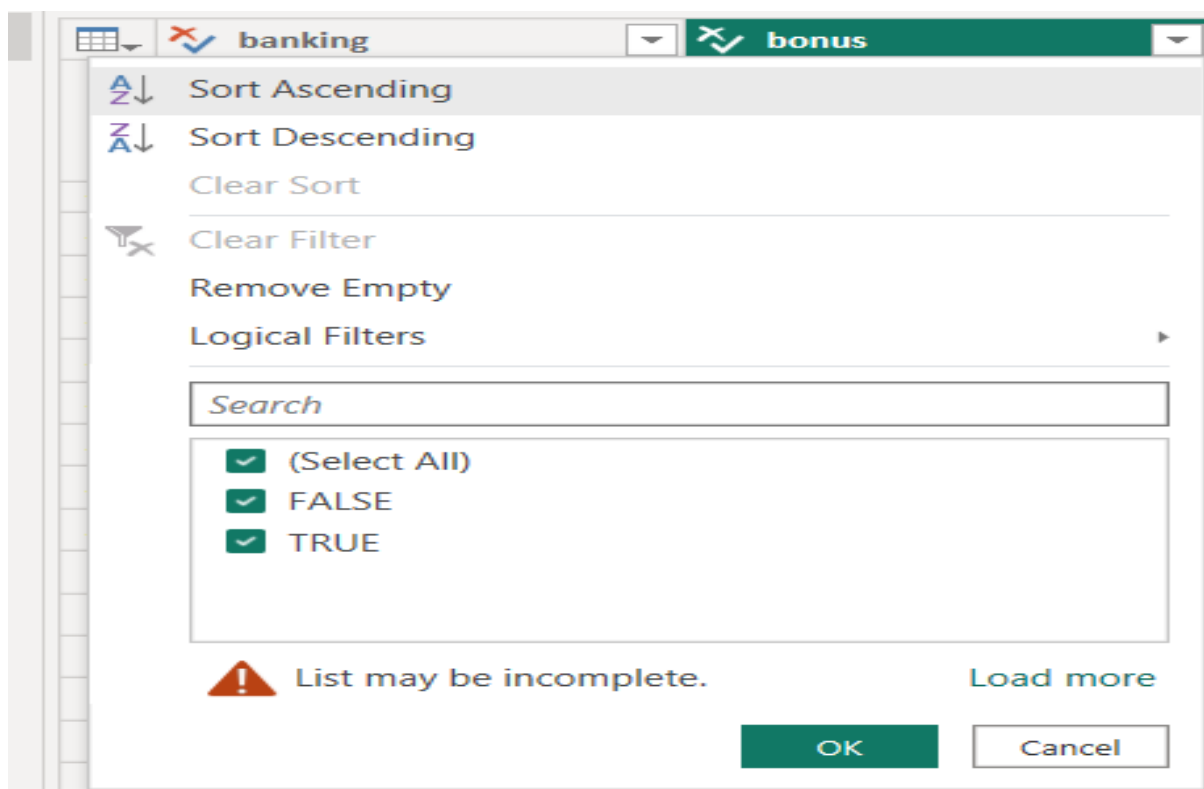
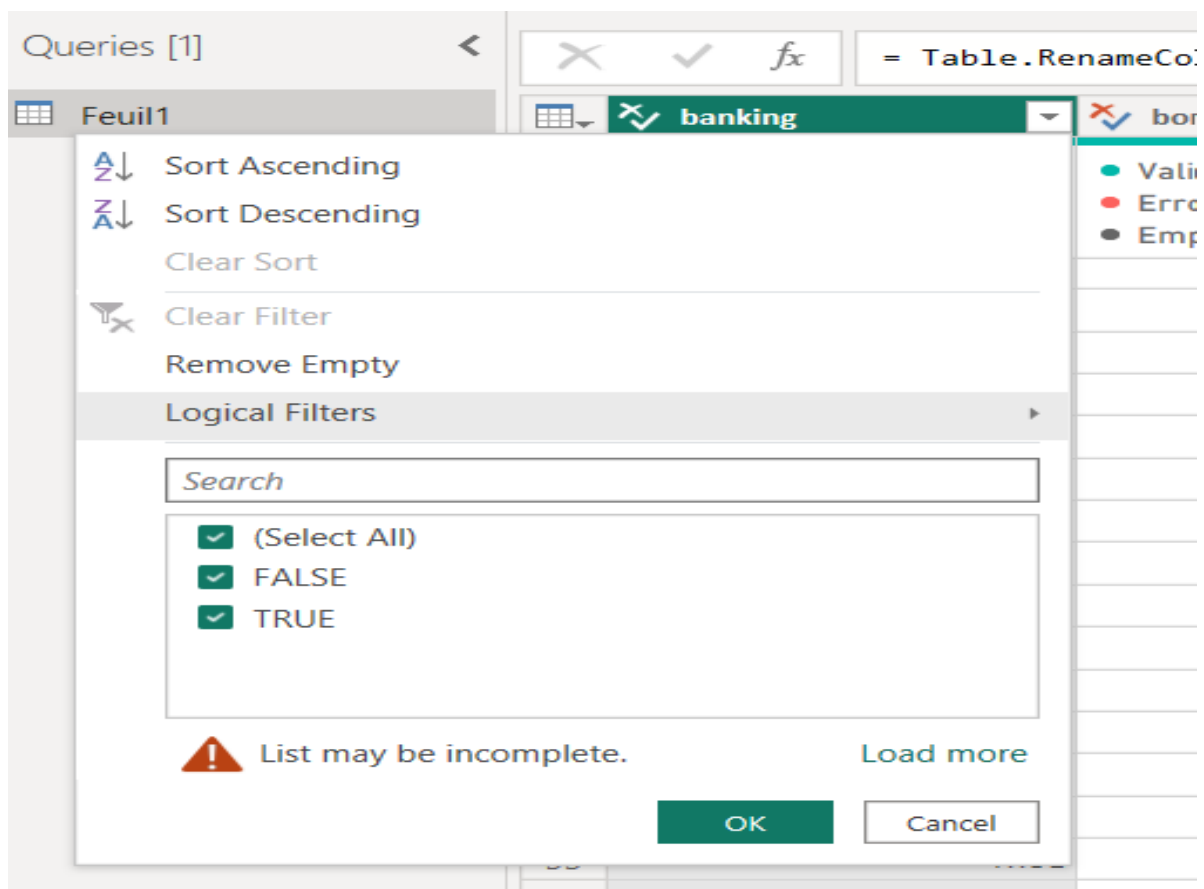
List may be incomplete.

Load more

OK

Cancel

STEP 7 — Check banking and bonus



STEP 8 — Create Update Date

Update Date	
● Valid	100%
● Error	0%
● Empty	0%
	02-06-2022
	06-04-2022
	27-09-2022
	08-06-2022
	23-05-2022
	13-09-2022
	06-09-2022
	14-09-2022
	12-09-2022
	03-10-2022
	13-09-2022
	26-09-2022

PROPERTIES

Name

 Feuill1

APPLIED STEPS

- Source
- Navigation
- Promoted Headers
- Changed Type
- Removed Duplicates
- Split Column by Delimiter
- Changed Type1
- Renamed Columns
- Inserted Date
- X Renamed Columns1**

STEP 9 — Create the Fact Table

Fact_BikeStation

Table: RenameColumns("Inserted Date",{"Date", "Update Date"})

	Contract Name	Bike Stands	Available Bike Stands	Available Bikes	Last Update	Update Date
100%	Valid 0% Error 0% Empty	100%	100%	100%	100%	100%
1	marseille	9	0	0	02-06-2022 15:14:38 +05:30	02-06-2022
2	marseille	13	0	0	06-04-2022 12:54:35 +05:30	06-04-2022
3	marseille	15	0	0	27-09-2022 10:57:48 +05:30	27-09-2022
4	marseille	15	15	0	08-06-2022 11:50:13 +05:30	08-06-2022
5	nantes	0	0	0	23-05-2022 20:11:50 +05:30	23-05-2022
6	marseille	10	0	0	13-09-2022 11:28:01 +05:30	13-09-2022
7	marseille	24	24	0	06-09-2022 18:02:17 +05:30	06-09-2022
8	marseille	20	13	7	14-09-2022 13:27:09 +05:30	14-09-2022
9	marseille	10	0	0	12-09-2022 11:18:28 +05:30	12-09-2022
10	marseille	20	2	0	03-10-2022 11:34:01 +05:30	03-10-2022
11	marseille	10	10	0	13-09-2022 15:41:24 +05:30	13-09-2022
12	marseille	14	0	0	26-09-2022 16:44:57 +05:30	26-09-2022

PROPERTIES

Name
Fact_BikeStation

APPLIED STEPS

Source
Navigation
Promoted Headers
Changed Type
Removed Duplicates
Split Column by Delimiter
Changed Type1
Renamed Columns
Inserted Date
Renamed Columns1

STEP 10 — Create Dim City

The screenshot displays the Power BI Desktop interface. On the left, the 'Fact_BikeStation' and 'Dim_City' tables are listed. The main area shows a data table with 18 rows, all containing the value 'marseille'. A dropdown menu for 'Contract Name' is open, showing 'Valid' (100%), 'Error' (0%), and 'Empty' (0%). The table is named 'Table.SelectColumns(Source,{'Contract Name'})'. On the right, the 'PROPERTIES' and 'APPLIED STEPS' sections are visible. The 'APPLIED STEPS' section shows 'Source' and 'Removed Other Columns'.

Contract Name

Valid

100%

Error

0%

Empty

0%

1

maribor

2

marseille

3

nantes

4

jcdecauxbike

5

nancy

6

rouen

7

lyon

8

santander

9

creteil

10

seville

11

besancon

12

namur

13

mulhouse

14

Bruxelles-Capitale

15

amiens

16

cernev-ontoise

PROPERTIES

Name

Dim_City

All Properties

APPLIED STEPS

Source

Removed Other Columns

Removed Duplicates

Fact_BikeStation

Dim_City

City

Valid

100%

Error

0%

Empty

0%

1

maribor

2

marseille

3

nantes

4

jcdecauxbike

5

nancy

6

rouen

7

lyon

8

santander

PROPERTIES

Name

Dim_City

All Properties

APPLIED STEPS

Source

Removed Other Columns

Removed Duplicates

Renamed Columns

STEP 11 — Create Dim_Station

Queries [3]

Fact_BikeStation

Dim_City

Dim_Station

Table.SelectColumns(Source,["number", "name", "address", "Latitude", "Longitude"])

number

name

address

Latitude

Longitude

Valid

100%

Error

0%

Empty

0%

Valid

100%

Error

0%

Empty

0%

Valid

100%

Error

0%

Empty

0%

Valid

100%

Error

0%

Empty

0%

Valid

100%

Error

0%

Empty

0%

1

2000

City Central Bike Station 1

102 Main Street, City

46.54887668

15.63502653

2

1297

City Central Bike Station 2

103 Main Street, City

43.29348574

5.3784238

3

8265

City Central Bike Station 3

104 Main Street, City

43.26931856

5.389387542

4

8065

City Central Bike Station 4

105 Main Street, City

43.26727252

5.372088524

5

1011

City Central Bike Station 5

106 Main Street, City

47.19504

-1.5575

6

4342

City Central Bike Station 6

107 Main Street, City

43.30599345

5.39367826

7

6176

City Central Bike Station 7

108 Main Street, City

43.28477598

5.385593764

8

3322

City Central Bike Station 8

109 Main Street, City

43.31233703

5.368741804

9

1264

City Central Bike Station 9

110 Main Street, City

43.29789631

5.381132216

10

1141

City Central Bike Station 10

111 Main Street, City

43.29630767

5.376020663

Query Settings

PROPERTIES

Name

Dim_Station

All Properties

APPLIED STEPS

Source

Removed Other Columns

Queries [3]

Fact_BikeStation

Dim_City

Dim_Station

Table.RenameColumns(#"Removed Other Columns",({"number", "Station_ID"}))

Station_ID

name

address

Latitude

Longitude

Valid

100%

Error

0%

Empty

0%

Valid

100%

Error

0%

Empty

0%

Valid

100%

Error

0%

Empty

0%

Valid

100%

Error

0%

Empty

0%

Valid

100%

Error

0%

Empty

0%

1

2000

City Central Bike Station 1

102 Main Street, City

46.54887668

15.63502653

2

1297

City Central Bike Station 2

103 Main Street, City

43.29348574

5.3784238

3

8265

City Central Bike Station 3

104 Main Street, City

43.26931856

5.389387542

4

8065

City Central Bike Station 4

105 Main Street, City

43.26727252

5.372088524

5

1011

City Central Bike Station 5

106 Main Street, City

47.19504

-1.5575

6

4342

City Central Bike Station 6

107 Main Street, City

43.30599345

5.39367826

7

6176

City Central Bike Station 7

108 Main Street, City

43.28477598

5.385593764

8

3322

City Central Bike Station 8

109 Main Street, City

43.31233703

5.368741804

9

1264

City Central Bike Station 9

110 Main Street, City

43.29789631

5.381132216

10

1141

City Central Bike Station 10

111 Main Street, City

43.29630767

5.376930662

11

5058

City Central Bike Station 11

112 Main Street, City

43.29280086

5.401285286

12

4299

City Central Bike Station 12

113 Main Street, City

43.29626341

5.406000803

13

1245

City Central Bike Station 13

114 Main Street, City

43.29902794

5.381500047

14

7153

City Central Bike Station 14

115 Main Street, City

43.28946761

5.363709908

15

6154

City Central Bike Station 15

116 Main Street, City

43.29032977

5.375548342

16

1002

City Central Bike Station 16

117 Main Street, City

43.29613878

5.370771385

17

1224

City Central Bike Station 17

118 Main Street, City

43.29582518

5.385996091

18

1287

City Central Bike Station 18

119 Main Street, City

43.29952098

5.385047449

Query Settings

PROPERTIES

Name

Dim_Station

All Properties

APPLIED STEPS

Source

Removed Other Columns

Renamed Columns

STEP 11A — Check Station_ID uniqueness

Queries [3]

Fact_BikeStation

Dim_City

Dim_Station

X

✓

fx

= List.NonNullCount(#"Renamed Columns"[Station_ID])

2856

Query Settings

PROPERTIES

Name

Dim_Station

All Properties

APPLIED STEPS

Source

Removed Other Columns

Renamed Columns

Calculated Count

STEP 11B — Create Station_Key

Custom Column

Add a column that is computed from the other columns.

New column name

Station_Key

Custom column formula ⓘ

= Text.From([Station_ID]) & "_" & [name]

Available columns

Station_ID

name

address

Latitude

Longitude

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK

Cancel

ABC123

Station_Key

Valid100%

Error0%

Empty0%

2000_City Central Bike Station 1

1297_City Central Bike Station 2

8265_City Central Bike Station 3

8065_City Central Bike Station 4

1011_City Central Bike Station 5

4342_City Central Bike Station 6

6176_City Central Bike Station 7

3322_City Central Bike Station 8

1264_City Central Bike Station 9

1191_City Central Bike Station 10

5058_City Central Bike Station 11

4293_City Central Bike Station 12

1245_City Central Bike Station 13

PROPERTIES

Name

Dim_Station

All Properties

APPLIED STEPS

Source

Removed Other Columns

Renamed Columns

Added Custom

STEP 12 — Create the same Station_Key in Fact_BikeStation

Custom Column

Add a column that is computed from the other columns.

New column name

Station_Key

Custom column formula

= Text.From([number]) & "_" & [name]

Available columns

- number
- name
- address
- Latitude
- Longitude
- banking
- bonus

<< Insert

[Learn about Power Query formulas](#)

✓ No syntax errors have been detected.

OK

Cancel

ABC123

Station_Key

Valid

100%

Error

0%

Empty

0%

2000_City Central Bike Station 1

1297_City Central Bike Station 2

8265_City Central Bike Station 3

8065_City Central Bike Station 4

1011_City Central Bike Station 5

4342_City Central Bike Station 6

6176_City Central Bike Station 7

3322_City Central Bike Station 8

1264_City Central Bike Station 9

1191_City Central Bike Station 10

5058_City Central Bike Station 11

4293_City Central Bike Station 12

1245_City Central Bike Station 13

7153_City Central Bike Station 14

PROPERTIES

Name

Fact_BikeStation

All Properties

APPLIED STEPS

Source

Navigation

Promoted Headers

Changed Type

Removed Duplicates

Split Column by Delimiter

Changed Type1

Renamed Columns

Inserted Date

Renamed Columns1

Added Custom

STEP 13 — Create Dim_Date

Table: Dim_Date

Date
02-06-2022
06-04-2022
27-09-2022
08-06-2022
23-05-2022
13-09-2022
06-09-2022
14-09-2022
12-09-2022
03-10-2022
26-09-2022
10-10-2022
17-10-2022

STEP 14 — Add Year

Table: Dim_Date

Date	Year
02-06-2022	2022
06-04-2022	2022
27-09-2022	2022
08-06-2022	2022
23-05-2022	2022
13-09-2022	2022
06-09-2022	2022
14-09-2022	2022
12-09-2022	2022
03-10-2022	2022
26-09-2022	2022
10-10-2022	2022
17-10-2022	2022
24-10-2022	2022
04-10-2022	2022
30-08-2022	2022
08-11-2022	2022
22-11-2022	2022
11-10-2022	2022
28-11-2022	2022
29-11-2022	2022
19-12-2022	2022
02-11-2022	2022
14-11-2022	2022
21-03-2023	2023

STEP 15 — Add Month

Table: Dim_Date

Date	Year	Month
02-06-2022	2022	June
06-04-2022	2022	April
27-09-2022	2022	September
08-06-2022	2022	June
23-05-2022	2022	May
13-09-2022	2022	September
06-09-2022	2022	September
14-09-2022	2022	September
12-09-2022	2022	September
03-10-2022	2022	October
26-09-2022	2022	September
10-10-2022	2022	October
17-10-2022	2022	October
24-10-2022	2022	October
04-10-2022	2022	October
30-08-2022	2022	August
08-11-2022	2022	November
22-11-2022	2022	November
11-10-2022	2022	October
28-11-2022	2022	November
29-11-2022	2022	November
19-12-2022	2022	December
02-11-2022	2022	November
14-11-2022	2022	November
21-03-2023	2023	March

STEP 15 — Add Month Number

Month Number"}}

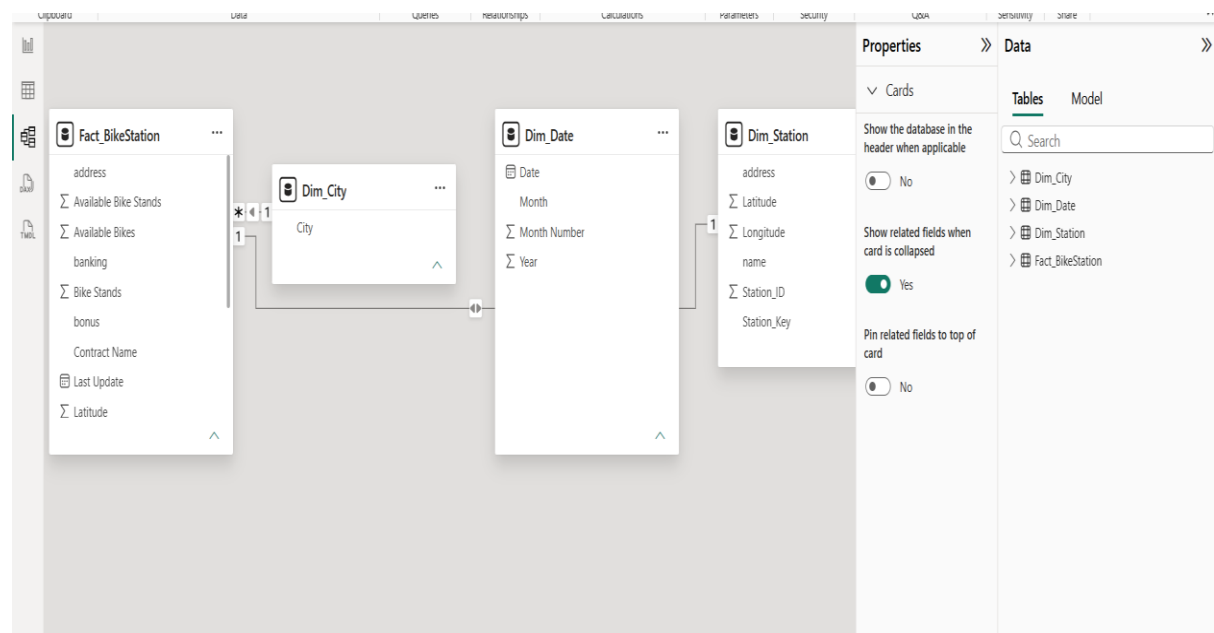
The screenshot shows the Power Query Editor. On the left, a table with 13 rows is displayed. The first row is the header 'Month Number'. The data rows contain values: 6, 4, 9, 6, 5, 9, 9, 9, 9, 10, 9, 10, 10, 10, 10, 8. To the right, the 'APPLIED STEPS' pane is visible, showing a list of steps: Source, Removed Other Columns, Removed Duplicates, Renamed Columns, Inserted Year, Inserted Month Name, Renamed Columns1, Inserted Month, and Renamed Columns2 (which is selected).

Month Number
6
4
9
6
5
9
9
9
9
10
9
10
10
10
10
8

STEP 16 — Finish Power Query

- Fact_BikeStation
- Dim_City
- Dim_Station
- Dim_Date

STEP 17 — Open Model View



STEP 17 — Verify the relationships

+ New relationship

Autodetect

Edit

Delete

Filter

☐	From: table (column) ↑	Relationship	To: table (column)	Status	
☐	Dim_Station (Station_Key)	1 ↔ 1	Fact_BikeStation (Station_Key)	Active	...
☐	Fact_BikeStation (Contract Na...	* ← 1	Dim_City (City)	Active	...

STEP 19 — Create Date Relationship

← New relationship

×

Select tables and columns that are related.

From table

Dim_Date

Date	Month	Month Number	Year
14 February 2...	February	2	2025
11 March 2025	March	3	2025
27 March 2025	March	3	2025

To table

Fact_BikeStation

	Longitude	name	number	Station_Key	status	Update Date
29548...	-5.974831157...	City Central Bi...	22	22_City Centr...	OPEN	12 November ...
27536...	-5.989037206...	City Central Bi...	19	19_City Centr...	OPEN	12 November ...
46	-5.971103	City Central Bi...	172	172_City Cent...	OPEN	12 November ...

Cardinality

One to many (1:*)

Cross-filter direction

Single

☒ Make this relationship active

☐ Apply security filter in both directions

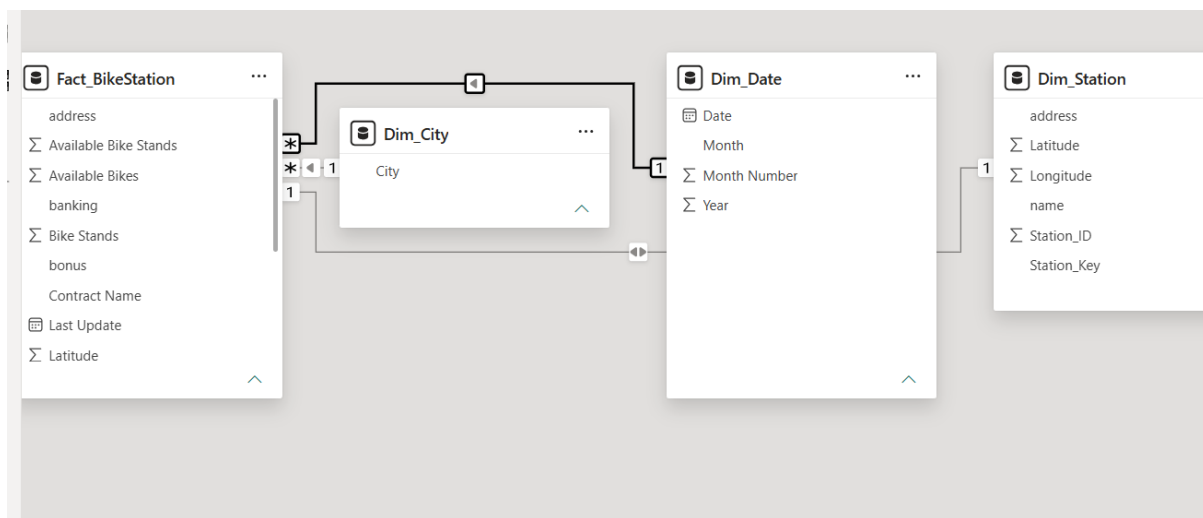
☐ Assume referential integrity

Save

Cancel

☐ From: table (column) ↑	Relationship	To: table (column)	Status	
☐ Dim_Station (Station_Key)	1 — 1	Fact_BikeStation (Station_Key)	Active	...
☐ Fact_BikeStation (Contract Na...	* — 1	Dim_City (City)	Active	...
☐ Fact_BikeStation (Update Date)	* — 1	Dim_Date (Date)	Active	...

STEP 20 — Close Manage Relationships



STEP 21 — DAX

Next we'll create the measures required.

But before creating DAX, we need to understand what dataset actually contains.

Dataset has:

- Bike Stands
- Available Bike Stands
- Available Bikes
- Status
- Banking
- Bonus
- City
- Date

- Latitude
- Longitude

It does not contain revenue, cost, or kilometres travelled. Therefore, we should not invent "revenue per km" calculations even though the assignment gives that as an example. We'll instead create meaningful bike-sharing performance measures from the fields actually available.

STEP 22 — Create DAX Measures

Measure 1 — Total Stations

The screenshot shows the Power BI Desktop interface. In the formula bar, the DAX measure is defined as:

```
1 Total Stations =
2 DISTINCTCOUNT(Fact_BikeStation[Station_Key])
```

The model view displays the following tables and their relationships:

- Fact_BikeStation**: Fact table with fields: address, Available Bike Stands, Available Bikes, banking, Bike Stands, bonus, Contract Name, Last Update, Latitude.
- Dim_City**: Dimension table with field: City.
- Dim_Date**: Dimension table with fields: Date, Month, Month Number, Year.
- Dim_Station**: Dimension table with fields: address, Latitude, Longitude, name, Station_ID, Station_Key.

Relationships are shown as follows:

- Fact_BikeStation (1) to Dim_City (4) on City.
- Fact_BikeStation (1) to Dim_Date (1) on Date.
- Fact_BikeStation (1) to Dim_Station (1) on Station_Key.

The Properties pane shows the measure name 'Total Stations' and its home table 'Fact_BikeStation'. The Data pane shows the Fact_BikeStation table with fields: address, Available Bike Stands, Available Bikes, banking, Bike Stands, bonus, Contract Name, Last Update, Latitude.

Measure 2 — Total Bike Capacity

The screenshot shows the Power BI Desktop interface. In the formula bar, the DAX measure is defined as:

```
1 Total Bike Capacity =
2 SUM(Fact_BikeStation[Bike Stands])
```

The model view displays the following tables and their relationships:

- Fact_BikeStation**: Fact table with fields: address, Available Bike Stands, Available Bikes, banking, Bike Stands, bonus, Contract Name, Last Update, Latitude.
- Dim_City**: Dimension table with field: City.
- Dim_Date**: Dimension table with fields: Date, Month, Month Number, Year.
- Dim_Station**: Dimension table with fields: address, Latitude, Longitude, name, Station_ID, Station_Key.

Relationships are shown as follows:

- Fact_BikeStation (1) to Dim_City (4) on City.
- Fact_BikeStation (1) to Dim_Date (1) on Date.
- Fact_BikeStation (1) to Dim_Station (1) on Station_Key.

The Properties pane shows the measure name 'Total Bike Capacity' and its home table 'Fact_BikeStation'. The Data pane shows the Fact_BikeStation table with fields: address, Available Bike Stands, Available Bikes, banking, Bike Stands, bonus, Contract Name, Last Update, Latitude.

Measure 3 — Total Available Bikes

1 Total Available Bikes =
2 SUM(Fact_BikeStation[Available Bikes])

Fact_BikeStation

- address
- Σ Available Bike Stands
- Σ Available Bikes
- banking
- Σ Bike Stands
- bonus
- Contract Name
- Last Update
- Latitude

Dim_City

- City

Dim_Date

- Date
- Month
- Σ Month Number
- Σ Year

Dim_Station

- address
- Σ Latitude
- Σ Longitude
- name
- Σ Station_ID
- Station_Key

Properties

General

Name: Total Available Bikes

Home table: Fact_BikeStation

Description: Enter a description

Create with Copilot

Content created by AI may not be accurate or appropriate, so review it carefully. [Read terms](#)

Synonyms: total available bikes

Display folder

Data

Tables

Search

- Dim_City
- Dim_Date
- Dim_Station
- Fact_BikeStation
 - address
 - Σ Available Bike Stands
 - Σ Available Bikes
 - banking
 - Σ Bike Stands
 - bonus
 - Contract Name
 - Last Update
 - Σ Latitude
 - Σ Longitude
 - name
 - Σ number
 - Station_Key
 - status
- Total Available Bikes
- Total Bike Capacity

Measure 4— Total Available Bike Stands

1 Total Available Bike Stands =
2 SUM(Fact_BikeStation[Available Bike Stands])

Fact_BikeStation

- address
- Σ Available Bike Stands
- Σ Available Bikes
- banking
- Σ Bike Stands
- bonus
- Contract Name
- Last Update
- Latitude

Dim_City

- City

Dim_Date

- Date
- Month
- Σ Month Number
- Σ Year

Dim_Station

- address
- Σ Latitude
- Σ Longitude
- name
- Σ Station_ID
- Station_Key

Properties

General

Name: Total Available Bike Stands

Home table: Fact_BikeStation

Description: Enter a description

Create with Copilot

Content created by AI may not be accurate or appropriate, so review it carefully. [Read terms](#)

Synonyms: total available bike stands

Display folder

Data

Tables

Search

- Dim_City
- Dim_Date
- Dim_Station
- Fact_BikeStation
 - address
 - Σ Available Bike Stands
 - Σ Available Bikes
 - banking
 - Σ Bike Stands
 - bonus
 - Contract Name
 - Last Update
 - Σ Latitude
 - Σ Longitude
 - name
 - Σ number
 - Station_Key
 - status
- Total Available Bike Stands

Measure 5— Bike Availability %

1 Bike Availability % =
2 DIVIDE([Total Available Bikes], [Total Bike Capacity], 0)

Fact_BikeStation

- address
- Σ Available Bike Stands
- Σ Available Bikes
- banking
- Σ Bike Stands
- bonus
- Contract Name
- Last Update
- Latitude

Dim_City

- City

Dim_Date

- Date
- Month
- Σ Month Number
- Σ Year

Dim_Station

- address
- Σ Latitude
- Σ Longitude
- name
- Σ Station_ID
- Station_Key

Properties

General

Name: Bike Availability %

Home table: Fact_BikeStation

Description: Enter a description

Create with Copilot

Content created by AI may not be accurate or appropriate, so review it carefully. [Read terms](#)

Synonyms: bike availability %

Display folder

Data

Tables

Search

- Dim_City
- Dim_Date
- Dim_Station
- Fact_BikeStation
 - address
 - Σ Available Bike Stands
 - Σ Available Bikes
 - banking
 - Σ Bike Stands
 - bonus
 - Contract Name
 - Last Update
 - Σ Latitude
 - Σ Longitude
 - name
 - Σ number
 - Station_Key
 - status

Measure 6— Create Station Utilization %

Measure Definition:

```
1 Station Utilization % =
2 DIVIDE(
3     [Total Bike Capacity] - [Total Available Bike Stands],
4     [Total Bike Capacity],
5     0
6 )
```

Model View:

- Fact_BikeStation** (Home table): address, Latitude, Longitude, name, Station_ID, Station_Key.
- Dim_City**: City.
- Dim_Date**: Date, Month, Month Number, Year.

Properties:

- Name:** Station Utilization %
- Home table:** Fact_BikeStation
- Description:** Enter a description
- Synonyms:** station utilization %

Data View:

- Tables:** Dim_City, Dim_Date, Dim_Station, Fact_BikeStation.
- Columns:** address, Available Bike Stands, Available Bikes, banking, Bike Stands, bonus, Contract Name, Last Update, Latitude.

Measure 7— Average Bikes per Station

Measure Definition:

```
1 Average Bikes per Station =
2 DIVIDE(
3     [Total Available Bikes],
4     [Total Stations],
5     0
6 )
```

Model View:

- Fact_BikeStation** (Home table): address, Latitude, Longitude, name, Station_ID, Station_Key.
- Dim_City**: City.
- Dim_Date**: Date, Month, Month Number, Year.

Properties:

- Name:** Average Bikes per Station
- Home table:** Fact_BikeStation
- Description:** Enter a description
- Synonyms:** average bikes per station, bikes, average bikes

Data View:

- Tables:** Dim_City, Dim_Date, Dim_Station, Fact_BikeStation.
- Columns:** address, Available Bike Stands, Available Bikes, banking, Bike Stands, bonus, Contract Name, Last Update, Latitude.

Measure 8— Average Station Capacity

Measure Definition:

```
1 Average Station Capacity =
2 DIVIDE(
3     [Total Bike Capacity],
4     [Total Stations],
5     0
6 )
```

Model View:

- Fact_BikeStation** (Home table): address, Latitude, Longitude, name, Station_ID, Station_Key.
- Dim_City**: City.
- Dim_Date**: Date, Month, Month Number, Year.

Properties:

- Name:** Average Station Capacity
- Home table:** Fact_BikeStation
- Description:** Enter a description
- Synonyms:** average station capacity, capacity, station capacity

Data View:

- Tables:** Dim_City, Dim_Date, Dim_Station, Fact_BikeStation.
- Columns:** address, Available Bike Stands, Available Bikes, banking, Bike Stands, bonus, Contract Name, Last Update, Latitude.

Measure 9— Format the percentage measures

The screenshot shows the Power BI Desktop interface. The DAX formula bar contains the following code:

```
1 Bike Availability % =  
2 DIVIDE(  
3 [Total Available Bikes],  
4 [Total Bike Capacity],  
5 0  
6 )
```

The model view shows a relationship between **Dim_City** and a date table. The **Dim_City** table has columns: address, Available Bike Stands, Available Bikes, banking, Bike Stands, bonus, Contract Name, Last Update, and Latitude. The date table has columns: Date, Month, Month Number, and Year. The **Properties** pane on the right shows the **Format** section for the selected measure:

- Format**: Percentage (dropdown)
- Percentage format**: Yes (toggle)
- Thousands separator**: No (toggle)
- Decimal places**: 1 (input field)

STEP 23 — Create the first KPI card

The screenshot shows the Power BI Desktop interface with the **Clipboard** pane on the left and the **Data** pane on the right. The **Clipboard** pane contains a KPI card with the following text:

3K
Total Stations

The **Data** pane shows the **Dim_City** table with columns: address, Available Bike Stands, Available Bikes, banking, Bike Stands, bonus, Contract Name, Last Update, and Latitude. The **Properties** pane on the right shows the **Format** section for the selected measure:

- Format**: Percentage (dropdown)
- Percentage format**: Yes (toggle)
- Thousands separator**: No (toggle)
- Decimal places**: 1 (input field)

STEP 24 — Total Bike Capacity Card

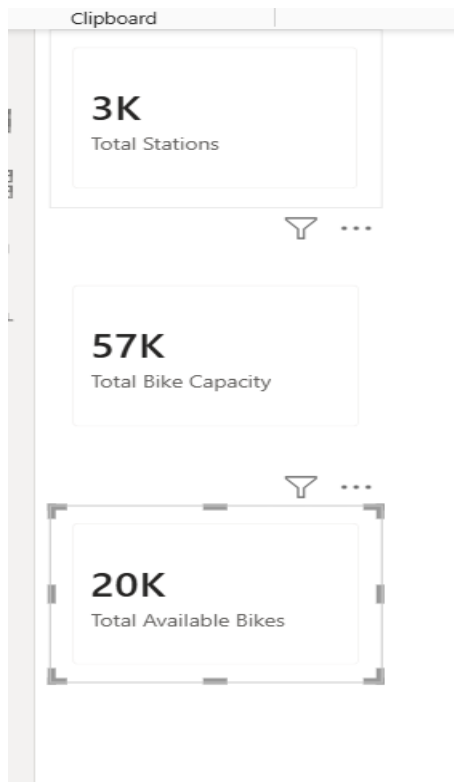
The screenshot shows the Power BI Desktop interface with the **Clipboard** pane on the left and the **Data** pane on the right. The **Clipboard** pane contains a KPI card with the following text:

3K
Total Stations

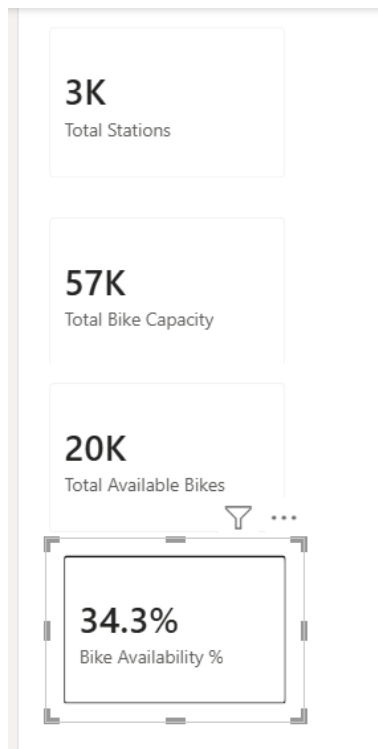
The **Data** pane shows the **Dim_City** table with columns: address, Available Bike Stands, Available Bikes, banking, Bike Stands, bonus, Contract Name, Last Update, and Latitude. The **Properties** pane on the right shows the **Format** section for the selected measure:

- Format**: Percentage (dropdown)
- Percentage format**: Yes (toggle)
- Thousands separator**: No (toggle)
- Decimal places**: 1 (input field)

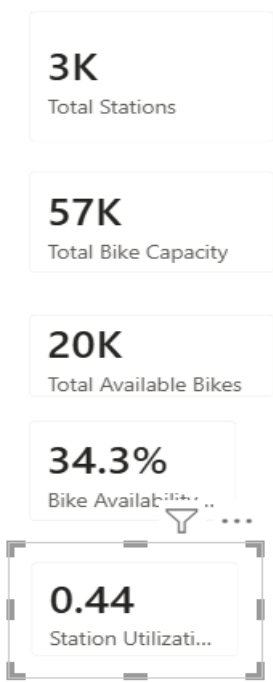
STEP 25 — Total Available Bikes



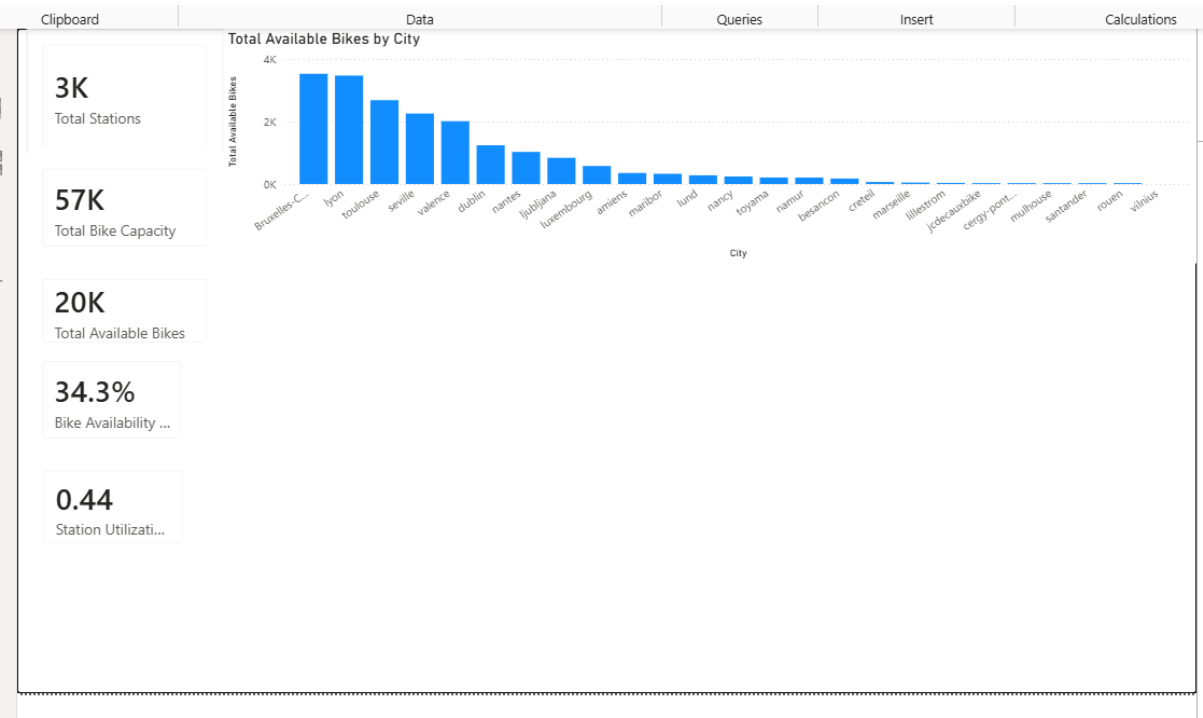
STEP 26 — Bike Availability %



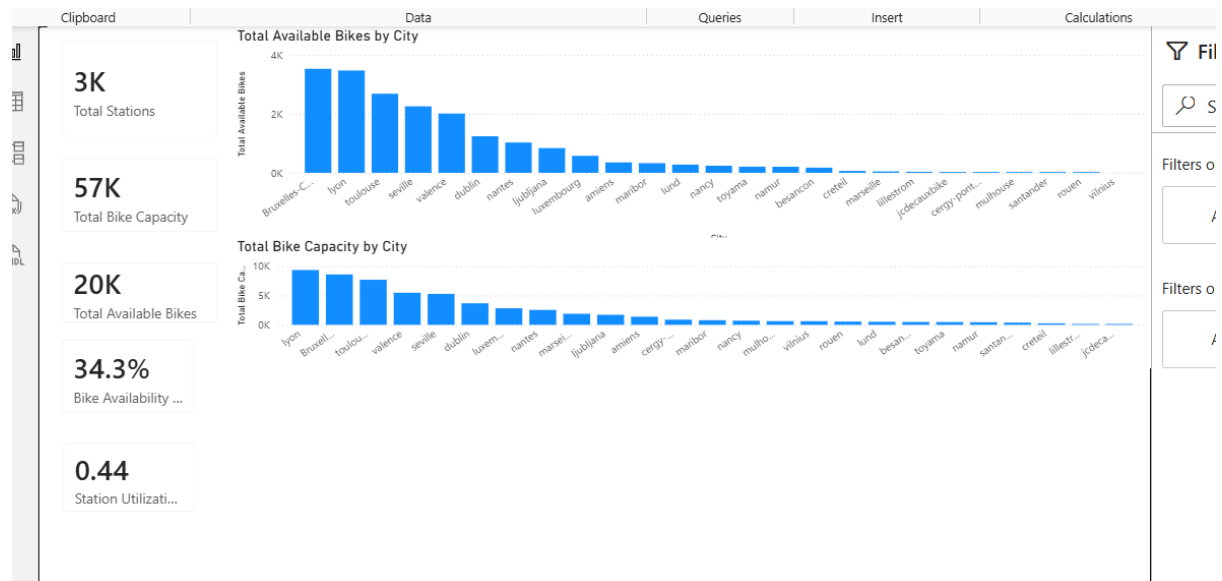
STEP 26 — Station Utilization %



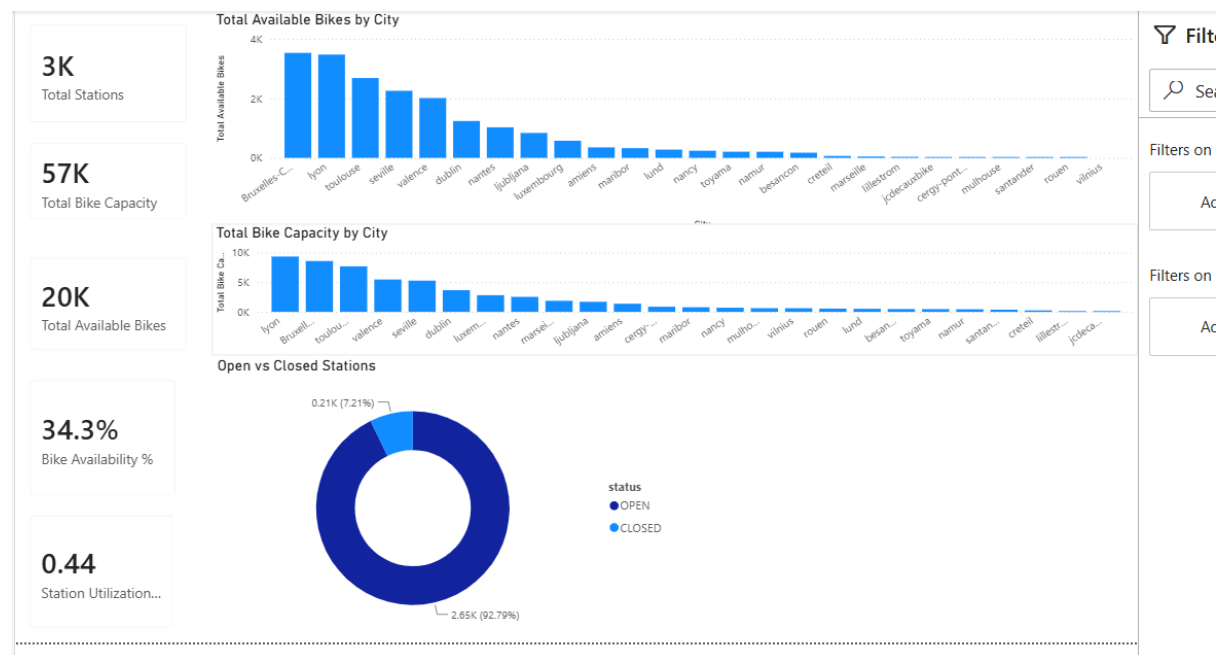
STEP 27 — First Chart: Bike Availability by City



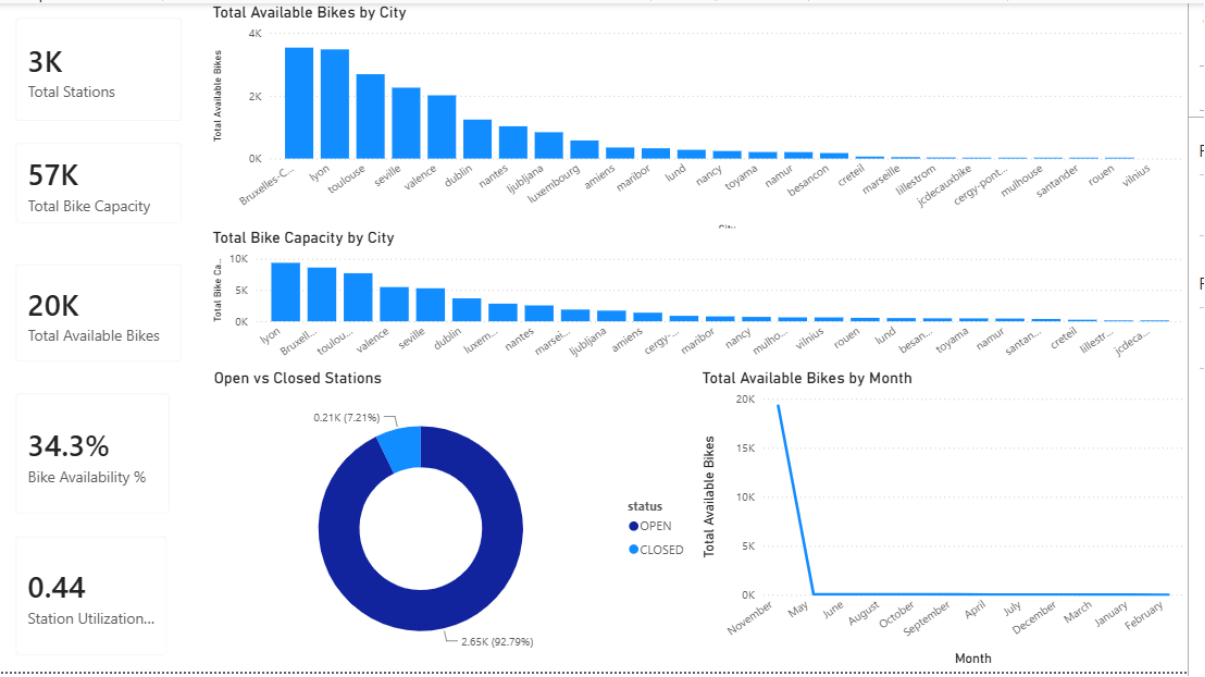
STEP 28 — Total Bike Capacity by City



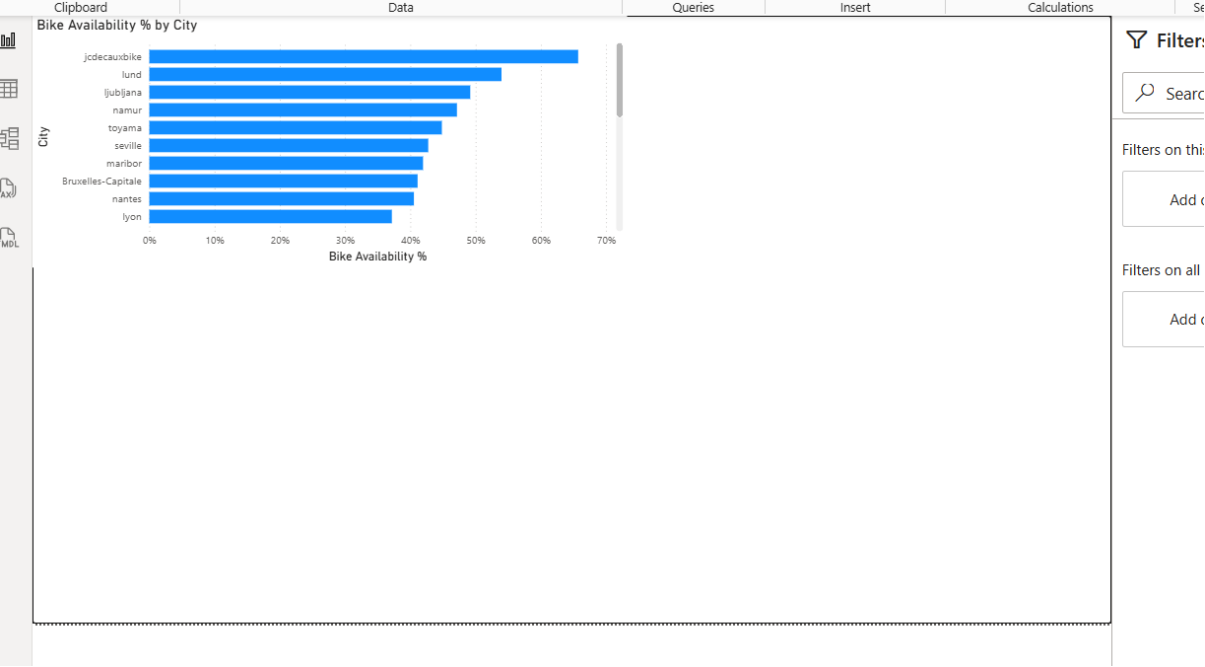
STEP 29 — Open vs Closed Stations



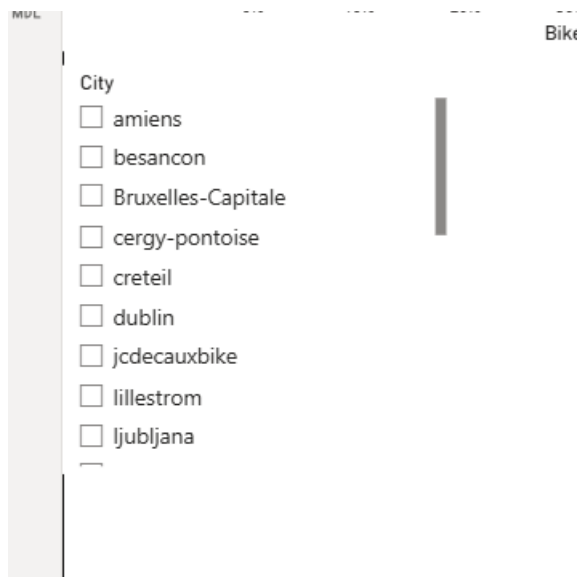
STEP 30 — Available Bikes by Month



STEP 31 — Bike Availability % by City



STEP 32 — Add City Slicer



STEP 33 — Add Year Slicer

